

LECTURE 4.1: VASCULAR SUPPLY OF THE BRAIN

ARTERIAL SUPPLY

- 18% of the total blood volume
- Loss of consciousness occurs in **less than 15secs** after blood flow to the brain has stopped; **irreparable damage occurs within 5mins**.

CIRCLE OF WILLIS - confluence of vessels that gives rise to all of the major cerebral arteries.

★ PRINCIPAL ARTERIES

- **INTERNAL CAROTID ARTERY** - enters brain from neck area; **divides common carotids and vertebral artery**.
- **VERTEBRAL ARTERY** - supplies **brain stem, cerebellum, occipital lobe, and parts of the thalamus**.
 - **Carotids** - normally supply the remainder of the forebrain.
- **ANTERIOR COMMUNICATING ARTERY AND ANTERIOR CEREBRAL ARTERY** - interconnects the carotid arteries.
- **POSTERIOR COMMUNICATING ARTERY** - connects the **2 posterior cerebral arteries**.

★ VERTEBROBASILAR ARTERY

- **BASILAR ARTERY** - 2 vertebral arteries fused after passing through the foramen magnum.
 - Terminates in the **interpeduncular cistern** in a *bifurcation* as the left and right posterior cerebral arteries.
- **POST. AND ANT. INFERIOR AND SUPERIOR CEREBELLAR ARTERIES** - several pairs of **small circumferential arteries** arise from **Basilar arteries**.
 - Post. inferior cerebellar arteries
 - Ant. inferior cerebellar arteries
 - Superior cerebellar arteries
 - Pontine branches
 - Internal auditory arteries
- **PENETRATING ARTERIES** - branches off the basilar artery and supplies the brain stem.

INTERNAL CAROTID ARTERY

- Passes through the **carotid canal of the skull**.
- Curves forward within the **cavernous sinus**.
- Up and backward through the dura.

★ Branches into:

- **MIDDLE CEREBRAL ARTERY** - larger
- **ANTERIOR CEREBRAL ARTERY** - smaller

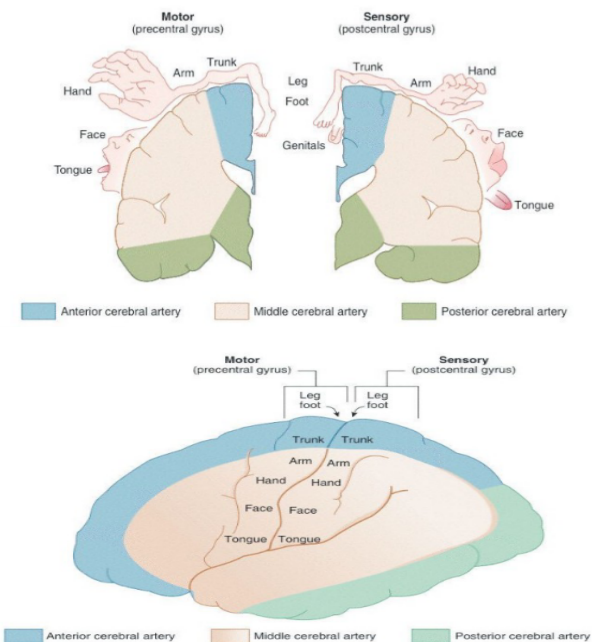
★ **OPHTHALMIC ARTERY** - first branch of the ICA

★ **INTERNAL CAROTID ARTERY**

- **ANTERIOR COMMUNICATING ARTERIES** - connects the **2 anterior cerebral arteries**.
 - ★ Forms an **ANASTOMOSIS** between left and right hemispheres (**important when one internal carotid becomes occluded**).
- **ANTERIOR CHOROIDAL ARTERY** - directly off the internal carotid
 - Carries blood **to the choroid plexus of the lateral ventricles** and adjacent brain structures.

★ CORTICAL SUPPLY

POSTERIOR CEREBRAL ARTERY	- Occipital lobe - Choroid plexuses of 3rd and lateral ventricles - Lower surface of the temporal lobe
ANTERIOR CEREBRAL ARTERY	- Medial surface of the frontal and parietal lobes, Basal ganglia, and anterior portion of the corpus callosum
MIDDLE CEREBRAL ARTERY	- Lateral surface of the frontal, parietal, and temporal portions of the brain - Basal ganglia and internal capsule



CEREBRAL BLOOD FLOW AND AUTOREGULATION

★ Pressure at the small cerebral arteries

- 450mmHg
- Ensures **adequate perfusion of the cerebral capillary beds** despite changes in systemic blood pressure.
- Increased activity in one cortical area is accompanied by a **shift in blood volume** to that area.

VENOUS DRAINAGE

★ Types of Channels:

- Dural venous sinus
- Dural meningeal veins
- Diploic Veins

INTERNAL DRAINAGE

- ★ **GREAT CEREBRAL VEIN OF GALEN** - drains interior of the cerebrum.
 - Drainage of the **Internal Cerebral Veins and Basal Veins of Rosenthal**
- ★ **STRAIGHT SINUS** - drains from the Great Cerebral Vein of Galen and inferior sagittal sinus.
- ★ **CAVERNOUS SINUS** - receive **venous drainage** from the superior and inferior ophthalmic veins and the superficial middle cerebral vein.

CORTIAL VEINS

Venous drainage - into the nearest large vein or sinus → confluence of the sinuses → **internal jugular vein**.

- **Division of veins of the cerebral convex surface:**
 - **Superior** - upward on the hemisphere's surface → superior sagittal sinus, **passing under any lateral lacunae**.
 - **Inferior** - end in the superficial middle cerebral vein. Inferior cerebral veins that do not end in this fashion **terminate in the transverse sinus**.

VENOUS SINUSES

INTRADURAL SINUSES - venous channels **lined by mesothelium** that lies between the inner and outer layers of the dura.

- **All sinuses drain into the internal jugular veins or pterygoid plexus.**

EMISSARY VEINS - connection of the **sinuses of the extracranial area**.

SUPERIOR SAGITTAL SINUS - between the **falx and the inside of the skull cap**.

INFERIOR SAGITTAL SINUS - in the **free edge of the falx**.

STRAIGHT SINUS - in the **seam between the falx and the tentorium**.

TRANSVERSE SINUSES - between **tentorium and its attachment on the skull cap**.

SIGMOID SINUSES - **S-curved** continuations of the transverse sinuses into the jugular veins; a **transverse and sigmoid sinus together form a lateral sinus**.

SPHENOPARIETAL SINUSES - drain the deep middle cerebral veins into the cavernous sinuses.

CAVERNOUS SINUSES (on the either side of the **sella turcica**) - receive drainage from multiple sources, including the ophthalmic and facial veins.

INFERIOR PETROSAL SINUSES - cavernous sinus → beginning of sigmoid sinus.

CEREBROVASCULAR DISORDERS

- **Most common cause of neurologic disability in adults** and the **3rd most common cause of death** in our society.
- **CNS is uniquely sensitive to ischemia** due to high metabolic rate and limited energy reserved.
- Ischemia results in rapid depletion of adenosine triphosphate (ATP) stores in the CNS.

- **Classification of Cerebrovascular Disease**
 - **Ischemic**
 - **Hemorrhagic**
- **TRANSIENT ISCHEMIA/TRANSIENT ISCHEMIC ATTACK** - brief episode of neurological dysfunction resulting from an **interruption in the blood supply to the brain/eye**.
- **PENUMBRA** - surrounds the area of the infarction.
 - Neurons may have been **metabolically compromised** and are **electrically silent but are not yet dead**.
 - Neurons may be salvageable.

OCCLUSIVE CEREBROVASCULAR DISEASE

- Insufficient blood supply to portions of the brain leads to **infarction and swelling with necrosis of brain tissue**.
- Can be caused by **Atherosclerosis (most common)**, **Thrombosis**, and **Embolism**.
- Extent of an infarct **depends on the presence/absence of adequate anastomotic channels**.
- Capillaries from adjacent vascular territories and corticomeningeal capillaries at the surface may reduce the size of the infarct.
- **Occlusion in the proximal of the Circle of Willis**
 - Collateral circulation through the **anterior communicating artery (ACA)** and **posterior communicating arteries (PCA)** may permit sufficient blood flow to prevent infarction.
- **Occlusion in the Internal Carotid Artery**
 - **Anastomotic flow in the retrograde direction** via the **ophthalmic artery**, from the external carotid artery, may provide adequate circulation thus preventing infarction.

ATHEROSCLEROSIS OF THE BRAIN

- Principal pathologic change in the arteries of the brain occurs in the **vasculature of the neck and brain**.
- **Disturbances in metabolism (fats)** - prominent associated change.
- **Hypertension** - accelerates the progression of **atherosclerosis** and is a treatable risk factor for stroke.
- **CAROTID BIFURCATION** - most common and most severe atherosclerosis locations:
 - Upper and lower parts of the basilar artery
 - Internal carotid artery at its trifurcation
 - The first third of the middle cerebral artery
 - The first part of the posterior cerebral artery.

TRANSIENT CEREBRAL ISCHEMIA

- Caused by **transient occlusion of an already narrow vessel**.
- Cause: **vasospasm**

- Full recovery in **less than 24 hours (commonly 30 minutes)**.
- *Result in reversible ischemic neurologic deficits, such as sudden vertigo or weakness, loss of cranial nerve function, or even brief loss of consciousness.*

LOCALIZATION OF THE VASCULAR LESION IN STROKE SYNDROMES

- **CAROTID ARTERY DISEASE** - often accompanied by **contralateral weakness or sensory loss**.
 - **Aphasia/apraxia** if the dominant hemisphere is involved.
 - Transient blurring/loss of vision (**amaurosis fugax**) may occur if there is retinal ischemia.
- After **occlusion of the ICA**:
 - Ischemia is often limited to the **territory of the middle cerebral artery**.
 - Weakness predominantly affects the **contralateral face and arm**.
 - The anterior and posterior cerebral artery territories are nourished via **collateral flow from the contralateral circulation** via the **anterior communicating and posterior communicating arteries**.
- **UNILATERAL OCCLUSION OF THE ANTERIOR CEREBRAL ARTERY** - weakness and sensory loss in the **contralateral leg**.
- **BILATERAL OCCLUSION OF THE ANTERIOR CEREBRAL ARTERY** - damage to **frontal lobes** → state of **akinetic mutism**: patient is indifferent and apathetic, moving little, and not speaking (even though there is no paralysis of the immobile limbs).
- **VERTEBROBASILAR ARTERY DISEASE** - presents with vertigo +
 - **Ataxia** (impaired coordination)
 - **Dysarthria** (slurred speech)
 - **Dysphasia** (impaired swallowing).
 - Nausea, and vomiting may be present
 - **Diplopia**: double vision (if oculomotor complex is involved)

HYPERTENSION HEMORRHAGE

- Caused by **chronic uncontrolled hypertension**
 - May result in the formation of **small areas of vessel distention—microaneurysm**—mostly in small arteries that arise from much larger vessels.
- Rupture of an aneurysm - **intracerebral hemorrhage**.
- **Most common sites of Intracerebral Hemorrhage** (in order of frequency)
 - **Lentiform nucleus** - specifically Putamen (**most common**).
 - **Thalamus**
 - **White matter of cerebral hemispheres**
 - **Pons**
 - **Cerebellum**

- Blood clot compresses and may destroy adjacent brain tissue.
 - Hemorrhages may compress the underlying **fourth ventricle and produce acute hydrocephalus**.

SUBARACHNOID HEMORRHAGE

- From **ruptured aneurysms or vascular malformations** (generally bleeds into the subarachnoid space).
- Vascular malformations (AVMs) often occur in **younger people** and are found on the brain surface.
- May be **congenital (berry aneurysm)** or the result of **infection**.
- Most frequent cause:
 - **Tearing of the bridging veins between brain surface and dural sinus**.
 - Can occur as a result of a relatively **minor trauma**.
- **VASCULAR SPASM** - complication that can lead to infarct.
- **CONGENITAL BERRY** - seen most frequently in the Circle of Willis or in the middle cerebral trifurcation.
 - **Common at sites of arterial branching**
 - Infrequently in vessels of the posterior fossa.

EPIDURAL HEMORRHAGE

- Most frequent cause:
 - Bleeding from a **torn meningeal vessel** (usually an artery) may lead to an **extradural** (outside the dura) accumulation of blood that can rapidly compress the brain, progressing to herniation or death.
- **Fracture of the skull** can cause this type of epidural/extradural hemorrhage.
- There is often **pulsating exophthalmos**
- **Interventional methods** - inserting a balloon/instrumentation into the shunt via a catheter or surgery may correct the problem.

ARTERIOVENOUS MALFORMATION SHUNTS

- **Cerebral arteries and veins form abnormal tangles/webs** (can occur as developmental anomalies).
- Some AVMs are clinically silent
- Trauma can also cause the rupture of adjacent vessels.